Are there any other symmetry principles having to do with space, time, and motion that the laws of nature should respect? If you think about this you might come up with one more possibility. The laws of physics should not care about the angle from which you make your observations. For instance, if you perform some experiment and then decide to rotate all of your equipment and do the experiment again, the same laws should apply. This is known as rotational symmetry, and it means that the laws of physics treat all possible orientations on equal footing. It is a symmetry principle that is on par with the previous ones discussed.

Are there others? Have we overlooked any symmetries? You might suggest the gauge symmetries associated with the nongravitational forces, as discussed in Chapter 5. These are certainly symmetries of nature, but they are of a more abstract sort; our focus here is on symmetries that have a direct link to space, time, or motion. With this stipulation, it's now likely that you can't think of any other possibilities. In fact, in 1967 physicists Sidney Coleman and Jeffrey Mandula were able to prove that no other symmetries associated with space, time, or motion could be combined with those just discussed and result in a theory bearing any resemblance to our world.

Subsequently, though, close examination of this theorem, based on insights of a number of physicists revealed precisely one subtle loophole: The Coleman-Mandula result did not exploit fully symmetries sensitive to something known as spin.

Spin

An elementary particle such as an electron can orbit an atomic nucleus in somewhat the same way that the earth orbits the sun. But, in the traditional point-particle description of an electron, it would appear that there is no analog of the earth's spinning around on its axis. When any object spins, points on the axis of rotation itself—like the central point of a spinning Frisbee—do not move. If something is truly pointlike, though, it has no "other points" that lie off of any purported spin axis. And so it would appear that there simply is no notion of a point object spinning. Many years ago, such reasoning fell prey to yet another quantum-mechanical surprise.

In 1925, the Dutch physicists George Uhlenbeck and Samuel Goudsmit realized that a wealth of puzzling data having to do with